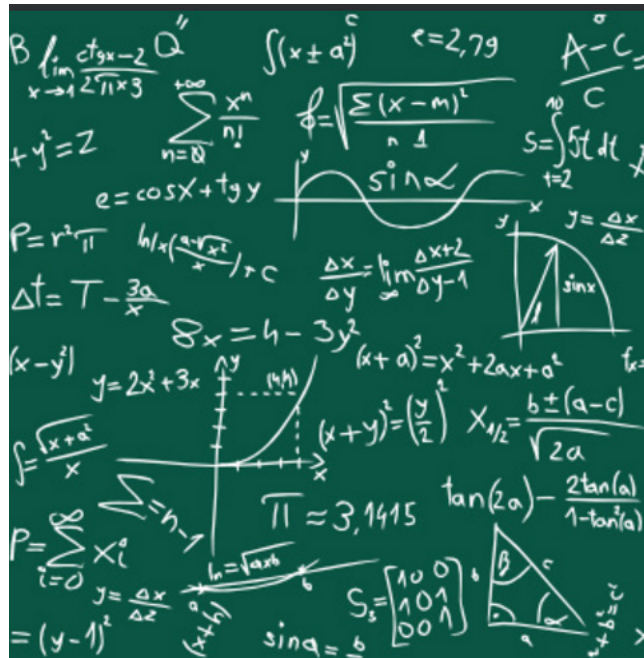


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Additional exercises



- **Authors:** Fabio Furini and Gianluca Priori.
- **Book:** “*Esercitazioni Analisi matematica 1*”, Marco Bramanti, Esculapio.

Exercise 1

Compute, if it exists, the limit of the sequence, justifying the steps taken:

$$\lim_{n \rightarrow +\infty} \frac{2^{-n} + 3n^3}{\log^6 n + 2 - n^5}$$

Exercise 2

Compute, if it exists, the limit of the sequence, justifying the steps taken:

$$\lim_{n \rightarrow +\infty} \frac{\log n + 3n^3 \log n}{2^{1/n} + n^5}$$

Exercise 3

Compute, if it exists, the limit of the sequence, justifying the steps taken:

$$\lim_{n \rightarrow +\infty} \frac{n^2 + \log^3 n}{n \log n + 1}$$

Exercise 4

Compute, if it exists, the limit of the sequence, justifying the steps taken:

$$\lim_{n \rightarrow +\infty} \frac{n^{2+\frac{1}{n}}}{1 + n^{3-\frac{2}{n}}}$$

Exercise 5

Compute, if it exists, the limit of the sequence, justifying the steps taken:

$$\lim_{n \rightarrow +\infty} \left(\frac{n}{n+1} \right)^{2n}$$

Exercise 6

Compute, if it exists, the limit of the sequence, justifying the steps taken:

$$\lim_{n \rightarrow +\infty} \left(\frac{n+2}{n+1} \right)^{n^2}$$

Exercise 7

Compute, if it exists, the limit of the sequence, justifying the steps taken:

$$\lim_{n \rightarrow +\infty} \left(1 + \frac{3}{n} \right)^{-n}$$

Exercise 8

Compute, if it exists, the limit of the sequence, justifying the steps taken:

$$\lim_{n \rightarrow +\infty} \left(1 - \frac{2}{n} \right)^n$$

Exercise 9

Compute, if it exists, the limit of the sequence, justifying the steps taken:

$$\lim_{n \rightarrow +\infty} \frac{n + \log(n^2) - 2^n}{(\log n)^3 + n^2}$$

Exercise 10

Compute, if it exists, the limit of the sequence, justifying the steps taken:

$$\lim_{n \rightarrow +\infty} \frac{8 + 2n - n^{4/3}}{n^{-1/3} + 3n - n^{5/3}}$$

Exercise 11

Compute, if it exists, the limit of the sequence, justifying the steps taken:

$$\lim_{n \rightarrow +\infty} n \left(\sqrt{n^2 + 3} - n \right)$$

Exercise 12

Compute, if it exists, the limit of the sequence, justifying the steps taken:

$$\lim_{n \rightarrow +\infty} \frac{n^2 + \log(5n^3) + (\log n)^2}{2(\log n)^3 + n \log n}$$

Exercise 13

Compute, if it exists, the limit of the sequence, justifying the steps taken:

$$\lim_{n \rightarrow +\infty} \frac{8 + 2n^2 - n^{4/3}}{n^{-1/3} + 3n^2 - n^{5/3}}$$

Exercise 14

Compute, if it exists, the limit of the sequence, justifying the steps taken:

$$\lim_{n \rightarrow +\infty} \left(\sqrt{n^4 + 1} - n \sqrt{n^2 + 1} \right)$$

Exercise 15

Compute, if it exists, the limit of the sequence, justifying the steps taken:

$$\lim_{n \rightarrow +\infty} \frac{n^2 + 3^n + 2^{-n}}{2^{2n} + (\log n)^3}$$

Exercise 16

Compute, if it exists, the limit of the sequence, justifying the steps taken:

$$\lim_{n \rightarrow +\infty} \frac{1}{n} (\sin^2 n) \cdot \sin(n^2)$$

Exercise 17

Compute, if it exists, the limit of the sequence, justifying the steps taken:

$$\lim_{n \rightarrow +\infty} \frac{\cos(\pi n)}{\pi n} + \log\left(\sin \frac{\pi}{n}\right)$$